

G H Raison College of Engineering & Management, Pune

(An Autonomous Institute affiliated to SP Pune University)

S.Y. B. Tech. /Semester-III /Div-A/2023-24

CYBER SECURITY

FORMAL LANGUAGES AND AUTOMATA (UCCSL203)

Questions	B L	CO	PO
Q1. Check whether the given grammar G is ambiguous or not. 1. $E \rightarrow E + E$ 2. $E \rightarrow E - E$ 3. $E \rightarrow id$	2	Co2	
Q2 Derive string "aabbabba" for leftmost and rightmost derivation using following context free Grammar $S \rightarrow aB bA$ $A \rightarrow a aS bAA$ $B \rightarrow b bS aBB$	3	CO 2	PO1
Q3 Define Application of Context Free Language.	2	CO 2	PO2
Q4. Check whether the given grammar G is ambiguous or not. 1. $S \rightarrow aSb SS$ 2. $S \rightarrow \epsilon$	3	CO 2	PO1
Q5 Design a PDA for accepting a language $\{wcw^r\}$.	2	CO 1	PO2
Q6 Define Ambiguity concept with an example.	2	CO 1	PO3
Q7 Construct following CFG to PDA. $S \rightarrow OS1 1SO \epsilon$	3	CO 1	PO1
Q8. Prove that following grammar is ambiguous and convert it into unambiguous grammar. $S \rightarrow S+S$ $S \rightarrow id$	3	CO 1	PO3
Q9 Differentiate between pushdown automata and finite Automata.	2	C1	PO3
Q10. Construct a Pushdown Automata that accept the language $L\{WCW^R\}$ over the alphabet a,b,c where $(W \geq 1)$	3	CO 1	PO2

Q11. Construct following CFG to PDA. $S \rightarrow aSb$ $S \rightarrow a \mid b \mid \epsilon$	3	CO 1	PO3
Q12. Explain Push Down Automata in Detail.	3	CO 1	PO2
Q13. Design a PDA for accepting a language $\{a^n b^{2n} \mid n \geq 1\}$.	3	CO 1	PO2
Q14. Define Context Free Grammar with seven tuples.	3	CO 1	PO1
Q15. Construct with following CFG to PDA. $S \rightarrow aSa, S \rightarrow aSa, S \rightarrow c$	2	CO 1	PO2
Q16. Construct the PDA equivalent to following grammar. $S \rightarrow aAA$ $A \rightarrow aS \mid bS \mid a$	3	CO 2	PO3
Q17. Derive the String 00101 for leftmost and rightmost derivation with the help of following CFG. 1. $S \rightarrow A \mid B$ $A \rightarrow 0A \mid \epsilon$ $B \rightarrow 0B \mid 1B \mid \epsilon$	2	CO 2	PO2
Q18. Construct the following CFG to PDA. $S \rightarrow OBB$ $B \rightarrow OS \mid 1S \mid O$	3	CO 2	PO2

Q19. Define LMD and RMD Derivation in Detail.	2	CO 2	PO1
Q20. Design a PDA for accepting a language $\{0^n 1^m 0^n \mid m, n \geq 1\}$.	3	CO 3	PO3

<i>Blooms Taxonomy level</i>	1: Remember, 2: Understand, 3: Apply, 4: Analyse, 5: Evaluate, 6: Create	
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